

Advanced Machine Learning

Lab 4 --- Module 2

Instructions on how to submit your work on Canvas

To convert your Notebook to an HTML file, follow these steps:

- Open the Notebook that you want to convert to an HTML file.
- Ensure that the notebook is saved with the latest changes.
- In the Notebook interface, navigate to File in the menu bar.
- Click on Download as and select HTML (.html) from the dropdown menu. This will initiate the conversion process.
- The Jupyter Notebook will be converted to an HTML file and automatically downloaded to the default download location on your computer.
- Locate the downloaded HTML file on your computer and verify that the conversion was successful.
- Optionally, open the HTML file in a web browser to view and verify the content.
- Upload your HTML file on Canvas.

Make sure to save your Notebook before attempting to convert it to an HTML file.

Exercise 1 (3 points)

A typical example in sparsity aware learning is the denoising problem. The problem in signal denoising is that instead of the actual signal samples, $\tilde{\mathbf{y}}$, a noisy version of the corresponding observations, $\mathbf{y}$, is available; that is, $\mathbf{y} = \tilde{\mathbf{y}} + \boldsymbol{\eta}$, where $\boldsymbol{\eta}$ is the vector of noise samples. Under the sparse modeling framework, the unknown signal $\tilde{\mathbf{y}}$ is modeled as a sparse representation in terms of a specific known dictionary $\boldsymbol{\Psi}$, that is, $\tilde{\mathbf{y}} = \boldsymbol{\Psi} \boldsymbol{\theta}$. Moreover, the dictionary is allowed to be redundant (over-complete). Then the denoising procedure is realized in two steps. First, an estimate of the sparse representation vector, $\boldsymbol{\theta}$, is obtained via any LASSO formulation,

$$\begin{aligned} \underset{\boldsymbol{\theta} \in \mathbb{R}^{\ell}}{\text{minimize}} \quad & \|\boldsymbol{\theta}\|_1 \quad \text{subject to} \quad \|\mathbf{y} - \boldsymbol{\Psi} \boldsymbol{\theta}\|_2 \leq \epsilon. \end{aligned}$$

Second, the estimate of the true signal is computed as $\hat{\mathbf{y}} = \mathbf{\Psi} \hat{\mathbf{\theta}}$.

For this lab, you have to denoise the blurred version of the given 256×256 image. First, extract from the image all the possible sliding patches of size 12×12 . Confirm that $(256 - 12 + 1)^2 = 60,025$ patches in total are obtained. Next, a dictionary in which all the patches are sparsely represented needs to be designed. Specifically, the dictionary atoms are going to be those corresponding to the two-dimensional redundant DCT transform, and are obtained as follows

1. Consider vectors $\mathbf{d}_i = [d_{i,1}, d_{i,2}, \dots, d_{i,12}]^T, i=0, \dots, 13$, being the sampled sinusoids of the form

$$d_{i,t+1} = \cos\left(\frac{t\pi i}{14}\right), t = 0, \dots, 11.$$

Then make (12×14) matrix $\mathbf{D}$, having as columns the vectors $\mathbf{d}_i$ normalized to unit norm; $\mathbf{D}$ resembles a redundant DCT matrix.

2. Construct the $(12^2 \times 14^2)$ dictionary $\mathbf{\Psi}$ according to $\mathbf{\Psi} = \mathbf{D} \otimes \mathbf{D}$, where $\otimes$ denoted Kronecker product. Built in this way, the resulting atoms correspond to atoms related to the overcomplete 2D-DCT transform.

Exercise 2 (2 points)

As a next step, denoise each image patch separately. In particular, assuming that $\mathbf{y}_i$ is the i th patch reshaped in column vector, estimate a sparse vector $\mathbf{\theta}_i$ in $\mathbb{R}^{196}$ and obtain the corresponding denoised vector as $\hat{\mathbf{y}}_i = \mathbf{\Psi} \hat{\mathbf{\theta}}_i$. Finally, average the values of the overlapping patches in order to form the full denoised image. Measure the performance of LASSO by evaluating the peak-to-noise ratio (PSNR) which is given by

$$\text{PSNR} = 20 \log_{10} \left(\frac{m_I}{\sqrt{\text{MSE}}} \right),$$

where m_I is the maximum pixel value of the image and $\text{MSE} = \frac{1}{N_p} \|\mathbf{I} - \tilde{\mathbf{I}}\|_F^2$, where $\mathbf{I}$ and $\tilde{\mathbf{I}}$ are the original and noisy image matrices, N_p is equal to the total number of pixels, and the Frobenius norm for matrices has been employed. Try different values for the regularization parameter and see what is the effect on PSNR.